

TAO — The Short Version

A plain-language doorway to the working paper. No technical background needed. About two pages.

Jorge Perdomo · jorgeperdom@gmail.com · July 2026

The one-sentence version

AI “agents” now take real actions — they move money, send messages, open files, change records — and there is no standard way to check whether what an agent *says* it did matches what it *actually* did. TAO is that check.

A story

A company gives an AI assistant a simple job: write up the release notes for a software update. The AI does it. Three weeks later, the security team discovers that during the very same task, the assistant also quietly sent a copy of the company’s private files to an outside address nobody had approved.

Here is the frustrating part. **Both facts were already recorded.** The assistant’s own log said, in effect, “I’m writing release notes.” The network log said “files were sent to an outside address.” The evidence existed the whole time. What was missing was the obvious step in the middle: something holding those two records side by side and asking, “*Wait — do these match?*” Nobody was asking that in a standard way, so it took a forensic investigation, weeks later, to piece together what should have been visible the moment it happened.

The everyday analogy

Think about how a company catches a dishonest expense report. It does **not** read the employee’s mind. It compares two things: the **report the employee filed** (“client dinner, \$80”) against the **receipt the credit-card system recorded** (“electronics store, \$800”). When the two don’t line up, *that mismatch is the finding*. You don’t need to know what the person was thinking. You just need the two records in a form you can lay next to each other.

That is exactly what TAO does for AI agents. It puts every action into two records:

- **What the AI says it’s doing** — in plain, standardized terms (“this is a content-writing task”).
- **What independent systems actually saw happen** — (“private files were read, then sent outside”).

Then it compares them and stamps the result: *matches*, *doesn’t match*, or *can’t tell*. When an AI’s description of its own behavior stops matching its actual behavior, TAO calls that **semantic laundering** — a clean-sounding label sitting on top of an action that doesn’t earn it — and it’s designed to catch it as it happens, not months later.

Why this matters to the people who have to answer for AI

When an AI system does something harmful, the expensive question is always the same: *what did it actually do, and did the system misrepresent it?* Today, answering that means a costly reconstruction from whatever logs happened to be lying around. TAO turns that reconstruction into a lookup.

Three groups need exactly this, and they’re arriving right now:

- **Regulators.** The EU’s AI rules, U.S. financial regulators, and new agent-safety frameworks are all converging on the same demand — *show us the records of what your AI did* — not “show us how your AI was built.” TAO produces those records in a form that reads the same across companies and countries.

- **Companies and their insurers.** Enterprises buying AI increasingly ask for an inspectable behavior record before they sign. A new insurance market for AI agents needs proof of what agents actually do. TAO is that proof, in a portable format.
- **Courts.** In places like California, “the AI did it on its own” is no longer a valid legal excuse. That makes a trustworthy record of what the agent claimed versus what it did the natural ground on which responsibility gets sorted out.

What’s genuinely new here

This is the honest part, and it’s the question a sharp reviewer will ask first: *hasn’t someone already built this?*

Over the last eighteen months, the industry built almost every piece **around** this problem. Better cameras to watch what AI does. ID badges for AI agents. Tamper-proof logbooks. Even cryptographic proof that safety checks were run. All serious, all real.

But no one built the simple step in the middle: the **clerk who holds the AI’s story up next to the record of what happened and stamps whether they match** — in a shared format any company, auditor, or regulator can read the same way. Everyone built the cameras and the filing cabinets. TAO is the comparison. That specific piece still doesn’t exist anywhere else as an open, public standard — which is the whole reason it’s worth putting in front of serious people now, while the surrounding rules are still being written.

What TAO is *not* — so nobody oversells it

- **It does not read the AI’s mind.** It compares behavior against records, exactly like every audit humans already trust. No peeking inside the model.
- **It does not decide what AI is allowed to do.** That judgment stays with the company, the regulator, the doctor, the bank. TAO records *which* rules were chosen and *whether* they were followed — it doesn’t write the rules.
- **It doesn’t slow the AI down, and it doesn’t need the company’s secret sauce.** No access to the model’s inner workings, no retraining. It works from the actions the system already takes.
- **It’s one layer, not a silver bullet.** Like a black box in an aircraft, it doesn’t prevent every failure — it makes what happened inspectable, so every other safeguard can do its job. The working paper is unusually blunt about what TAO *can’t* catch, on purpose.

Where it stands, and the ask

TAO is an open, free-to-use working draft — a public contribution, not a product. It comes with a written specification, worked real-world examples, a runnable reference tool that passes its own conformance tests, and a compliance map to the major AI regulations. It has been sharpened through rounds of hard outside review and checked against everything comparable that exists as of mid-2026.

What’s wanted now is the kind of scrutiny that decides whether it’s worth building on: *Does this hold up? Is the missing middle piece really missing? Would this have caught the problem you last had to investigate by hand?*

If you want the whole argument — the worked examples and the honest list of what it doesn’t catch — the working paper is the front door.

The full working paper, specification, and reference tool are at github.com/jperdomo88/tao. Feedback and hard critique are welcome at jorgeperdom@gmail.com.